

Polynomial Chaos Expansion for uncertainty quantification in tunnel engineering analysis

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ABSTRACT: The paper focuses on uncertainty quantification of a tunnel using a surrogate model in the form of Polynomial Chaos Expansion (PCE). The tunnel case under investigation is the underground Oakwood Station - part of the Eglinton Crosstown Light Rail Transit (ECLRT). The 120 m long station comprises a 19 m wide by 15 m high cavern, housing a central platform and an upper concourse level, as well as under-platform ventilation. The tunnels were constructed in glacial soft ground deposits using the Sequential Excavation Method (SEM), with a low cover of approximately 11 m. Tunnel design was a complex task influenced by a wide range of load and material uncertainties, including soil properties and shotcrete material parameters. To address these uncertainties, a combination of the finite element method and polynomial chaos expansion for uncertainty quantification was utilized. Although various advanced adaptive techniques for the construction of PCEs exist, a key limitation is that a standard PCE typically approximates only a single random variable. In the case of complex analyses involving many quantities of interest or complete stress fields, constructing separate advanced surrogates one-by-one for each target variable becomes computationally demanding. This paper presents an application of a recently proposed algorithm for the adaptive construction of PCEs for vector-valued responses, resulting in a dramatic reduction in computational cost and thus significantly enhancing the scalability of existing PCE algorithms.

1 INTRODUCTION

With the rapid advancement of computational technology and the increasing demand for cost-effective yet sophisticated design solutions in the construction industry, numerical modelling has become the preferred approach across various structural and heavy civil engineering disciplines, including tunnel engineering. Finite Element (FE) methods, in particular, have proven to be indispensable tools for analysing complex underground structures, enabling the use of advanced constitutive models and detailed geometrical representations. Moreover, numerical modelling allows for the analysis of intricate geometries, including non-circular cross-sections, irregular interfaces, and complex junctions. In tunnel design and assessment, significant uncertainties arise from knowledge gaps, inherent variability in soil and rock

behaviour, structural properties, and their interactions, as well as from mathematical and model-based approximations. Previous research has demonstrated that incorporating probabilistic approaches to tunnel design can enable substantial optimisation with respect to the tunnel concrete linings’ structural reliability (Spyridis et al. 2016) or service life (Spyridis 2014), further enabling dramatic improvements in overall “sustainability by design” (Spyridis & Bergmeister 2024). This study continues the exploration of advanced analysis methods in tunnelling, with a particular focus on the application of surrogate modelling techniques, specifically, Polynomial Chaos Expansion (PCE) for uncertainty quantification. Comparative analyses are presented using the underground Oakwood Station, a section of the Eglinton Crosstown Light Rail Transit (ECLRT) project (Laubbichler et al. 2021). The 120 m-long station features a cavern measuring 19 m in width and 15 m in height. The results obtained from surrogate modelling are compared with those derived from the standard Monte Carlo sampling approach using the original FE model as the reference solution.

2 TUNNEL STRUCTURE

2.1 Description of analysed structure

The analysed cavern, one of the largest ever in soft ground, was constructed in glacial deposits using the SEM. The twin segmental running tunnels were excavated first, then partially back-filled within the footprint of the station, and the cavern was subsequently constructed removing the segments. A double side drift excavation sequence was chosen due to the presence of the segmental tunnels and the large size of the cavern. The initial outer lining was 45cm thick, steel fibre reinforced shotcrete and the two side walls 30cm thick.

The main geotechnical units are shown in Figure 1. The top fill was about 4m thick, followed by 8m of plastic till. Most of the tunnel was excavated in the sandy deposits, which were about 20m thick, underlain by clay.

2.2 Finite element model

A 2D plane strain model in software Abaqus was used to simulate the soil-structure interaction and staged excavation effects. A total of 1000 FE analyses were executed and the outputs included lining hoop forces, shear forces and bending moments as well as surface displacements. The main analysis steps are shown in Figure 2. Each step was preceded by a relaxation step to simulate the ground pre-relaxation, using the stiffness reduction method (Potts & Zdravkovic 2001).

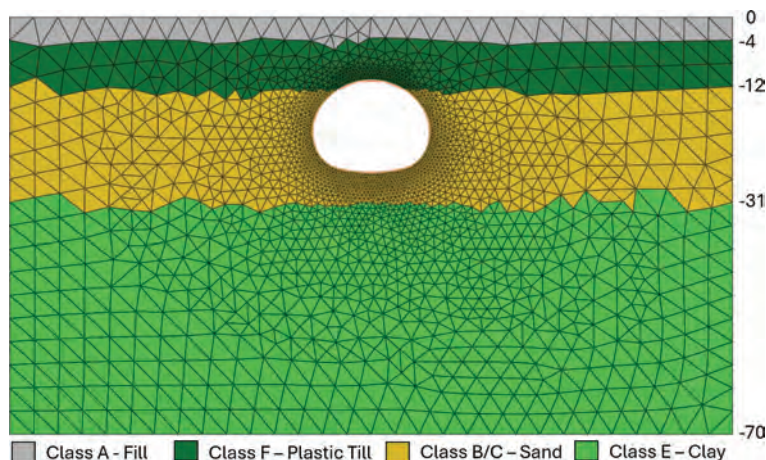


Figure 1. Geotechnical units simulated in the FE analysis.

The lining was simulated with beam elements, with a linear elastic model. The mean stiffness was taken as 13 GPa to account for the shotcrete strength/stiffness gain with time (John & Mattle 2003).

The soil was simulated with solid elements with a hardening Mohr-Coulomb material model. The parameters of class F and B/C units were considered more influential, and they were varied in the analyses, whereas the parameters of Class A and E2 remained unchanged. The soil parameters that varied were the cohesion c , friction angle ϕ , secant modulus E_{50} , unloading/reloading modulus E and the coefficient of lateral pressures k_0 . The mean values of these parameters are shown in Table 1, where index 1 denotes Class F and index 2 class B/C.

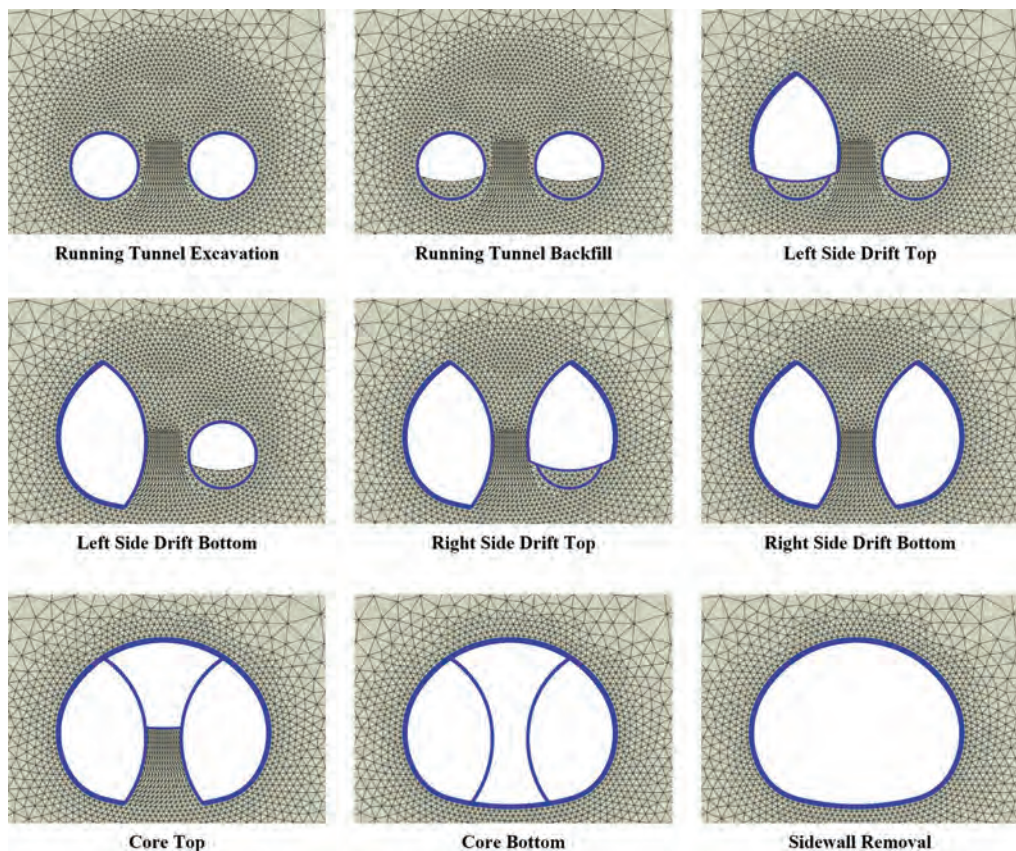


Figure 2. Main excavation steps simulated in the FE analysis.

3 UNCERTAINTY QUANTIFICATION

3.1 Polynomial chaos expansion

Considering a model response $Y = f(\mathbf{X})$ to be a random variable referenced as quantity of interest (QoI) with finite variance σ^2 , the polynomial chaos expansion of QoI is in the following form according to Ghanem & Spanos (1991):

$$Y = f(\mathbf{X}) = \sum_{\alpha \in N^M} \beta_{\alpha} \Psi_{\alpha}(\xi), \quad (1)$$

where M is the number of input random variables, $\beta\alpha$ are unknown deterministic coefficients and $\psi\alpha$ are multivariate basis functions orthonormal with respect to the joint probability density function (PDF) of ξ . The basis functions must be selected in dependence to probability distributions of components of the input random vector $\mathbf{X}$ which must be transformed to the associated standardized variables ξ as shown by Xiu & Karniadakis (2002).

Further, it is necessary to truncate PCE to a finite number of terms P . Truncated set of basis functions $A^{M,p}$ is typically dependent on given maximal polynomial order p and M as follows:

$$A^{M,p} = \left\{ \alpha \in N^M : |\alpha| = \sum_{i=1}^M \alpha_i \leq p \right\} \quad (2)$$

Deterministic coefficients $\beta\alpha$ can be obtained by intrusive and non-intrusive approaches. Non-intrusive methods utilize the original mathematical model (e.g. FE) as a black-box, which allows for their easy applications in combination with commercial software and thus is typically preferred for industrial applications. The most popular type of the non-intrusive approach is based on linear regression, i.e. ordinary least squares:

$$\beta = (\Psi^T \Psi)^{-1} \Psi^T \mathbf{y} \quad (3)$$

The original mathematical model must be evaluated to obtain a vector of results corresponding to generated sample points. Once the basis functions are created and experimental design (ED) is prepared, PCE coefficients can be estimated by OLS regression method. Unfortunately, truncated PCE solved by OLS is not highly computationally efficient and cannot be employed for practical examples with large number of input random variables due to the curse of dimensionality. The solution is an additional reduction of the truncated basis set by any model-selection algorithm such as Least Angle Regression (LAR) proposed by Efron et al. (2004). LAR automatically detects the most important basis functions for given ED resulting in sparse PCE. In this paper, implementation of an algorithm for construction of non-intrusive sparse PCE based on LAR is employed for numerical examples as implemented in UQPy (Tsapetis et al. 2023). Powerful post-processing is main advantage of PCE over other types of surrogate models. First of all, thanks to the explicit form of PCE it is possible to obtain a leave-one-out error Q^2 analytically without additional computational demands which can be further used for adaptive construction of PCE approximation. Besides the analytical derivation of Q^2 , it is possible to derive also the first four statistical moments directly from deterministic coefficients and exploit the PCE for global sensitivity analysis as shown by Novak (2022) and Novak et al. (2025).

However, constructing a large number of surrogate models individually is not scalable for high-dimensional quantities of interest (QoIs), such as 3D stress fields in structures. Therefore, we also investigate the potential of the recently proposed SAPCE algorithm by Loukrezis et al. (2025). The employed SAPCE is based on simple rationale exploiting structure of Eq. 3. Note that in typical settings, $\mathbf{y}$ represents a vector containing results obtained from FE models for given ED. However, if design matrix Ψ contains an identical set of basis functions, $\mathbf{y}$ can be extended to a matrix whose columns represent all QoIs, allowing Eq. (3) to be solved simultaneously for all investigated QoIs. The set of basis functions optimal for all QoIs can then be identified iteratively using average Sobol indices derived from the deterministic coefficients β . This approach results in a significant reduction of computational cost for high-dimensional QoIs. In this study, SAPCE reduced the computational cost by approximately one order of magnitude.

3.2 Methodology

The study focuses on the application of recently developed advanced algorithms for constructing Polynomial Chaos Expansions (PCE) in a real-world engineering problem — the

underground Oakwood Station, which is part of the Eglinton Crosstown Light Rail Transit (ECLRT) project. The ultimate objective is to perform uncertainty quantification (UQ), including sensitivity analysis of the structural response and selected surface locations. The FE model of the concrete lining comprises 113 elements, with three quantities of interest (QoIs) defined for each element — the internal forces N , V , and bending moments M . In addition, 23 vertical and horizontal displacements U of the tunnel and surface were selected for approximation. In total, 362 surrogate models were constructed to approximate all QoIs, including the internal forces and deformations of the tunnel and surrounding soil.

Two types of algorithms were employed to construct PCE approximations for each QoI using an identical experimental design (ED) consisting of 75 samples generated via Latin Hypercube Sampling (LHS). In the standard state-of-the-art approach, each PCE is constructed using the adaptive sparse algorithm implemented in the UQPy Python package. The maximum polynomial order is automatically selected as $p \in [3, 7]$ and the sparse basis set is determined using Least Angle Regression (LAR). The second algorithm employed is the recently proposed SAPCE method, developed for efficient approximation of high-dimensional QoIs. The results obtained from both approaches are compared in terms of accuracy, evaluated using a validation set of 1000 samples containing the corresponding results of the original FE model.

The stochastic model of the analysed structure includes 12 uncertain parameters related to the soil and structural properties. The assumed probability distributions of these parameters are summarized in the following table. Note that we assume uncertain parameters associated to concrete and both layers of soil.

Table 1. Stochastic model of the analysed structure.

Parameter	Distribution	Mean /lower bound	Standard deviation / upper bound
E_c [GPa]	Log-Normal	13	1
Relaxation [%]	Uniform	30	70
E/E_{50_1} [-]	Uniform	2.5	0.25
E_{50_1} [kPa]	Normal	65	5
c_1 [kPa]	Normal	30	5
ϕ_1 [deg]	Normal	30	1
k_{θ_1} [-]	Uniform	0.6	1.1
E/E_{50_2} [-]	Uniform	2.5	0.25
E_{50_2} [kPa]	Normal	130	10
c_2 [kPa]	Normal	5	1
ϕ_2 [deg]	Normal	42	1
k_{θ_2} [-]	Uniform	0.45	0.65

4 RESULTS AND DISCUSSION

First, it is necessary to examine the convergence rate of both algorithms with respect to the number of training samples, as illustrated in Figure 3. It can be observed that the standard PCE approach begins to converge for a minimum of 50 simulations, whereas SAPCE achieves meaningful results already with approximately 20 samples, although these results may be influenced by the random selection of samples in the experimental design (ED). Therefore, for the subsequent UQ analysis, an ED containing 75 samples was adopted, representing a robust design with low variability in terms of achieved accuracy.

Once the surrogate models are constructed, the first four statistical moments and Sobol' indices can be obtained analytically without any additional computational cost. The mean values and $\pm 3\sigma$ intervals of the internal forces are shown in the following figures. Although the influence of variance on the normal and shear internal forces is relatively small, in the case of bending moments there exists a critical location where the uncertainty of input variables results in an interval encompassing both negative and positive values. As shown, the

reference values obtained from 1000 validation samples are nearly identical for both methods in the case of internal forces. It can therefore be concluded that both surrogate models constructed with 75 samples are sufficiently accurate, with errors below 5% for most elements.

The approximation of surface deformations at selected locations above the tunnel is presented in Figure 4. Both techniques provide accurate estimates of the statistical moments compared to the reference FE solutions, although SAPCE slightly underestimates the variance

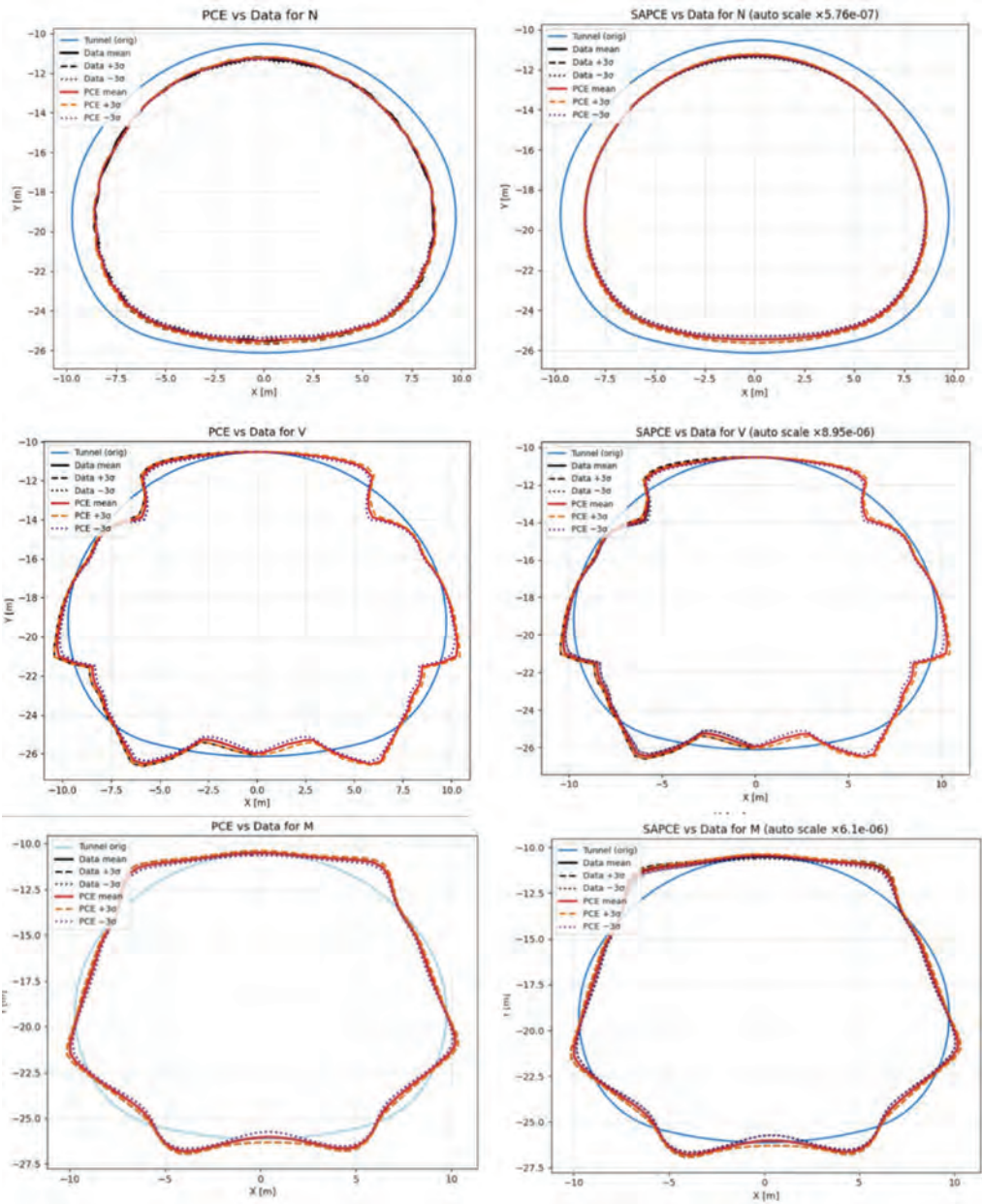


Figure 3. UQ of internal forces N (top), V (middle) and bending moments (bottom) approximated by standard PCE (left) and SAPCE (right). Red colour represents approximation, and black colour shows the reference solution obtained from validation set.

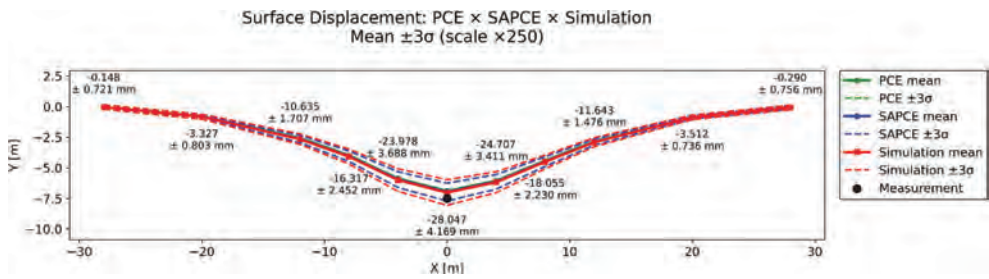


Figure 4. Surface displacement approximated by PCE (green) and SAPCE (blue) together with reference solution (red) and measurement (black dot).

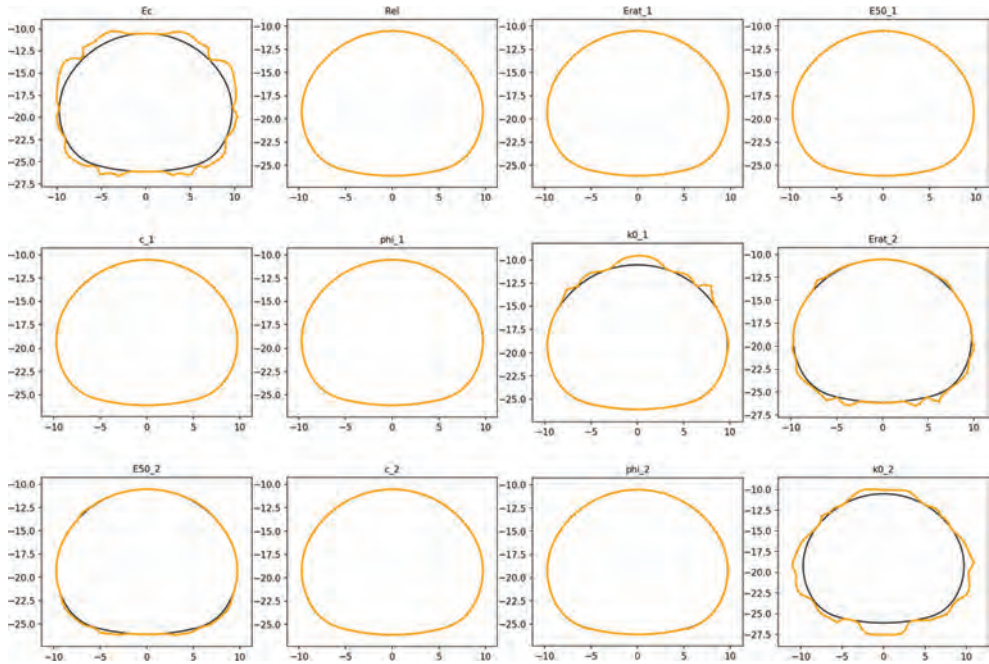


Figure 5. Sobol' indices of bending moments obtained from PCE.

and thus the $\pm 3\sigma$ intervals. The black dot in the figure represents the real-life measurement of displacement, which shows very good agreement with the estimated values.

Finally, the obtained total-order Sobol' indices for the bending moments are shown in Figure 5. Both methods yield nearly identical results. Interestingly, in most locations, the dominant input variables influencing the uncertainty of the QoIs are the elastic modulus E_c and the coefficient of earth pressure at rest k_0 for both soil layers. Similar findings are observed for the normal and shear internal forces. The results of the sensitivity analysis can thus be further exploited to reduce the stochastic model, significantly improving the computational efficiency of both approaches.

5 CONCLUSIONS

This paper presents the application of Polynomial Chaos Expansion for uncertainty quantification of a real-world engineering structure. The case study was underground Oakwood Station, which forms part of the Eglinton Crosstown Light Rail Transit project.

In addition to the standard approach, where adaptive sparse PCEs are constructed individually for each quantity of interest, it was demonstrated that the recently proposed SAPCE

algorithm for vector-valued QoIs can dramatically reduce the overall computational cost of the process. Once the PCEs were constructed for all 362 QoIs, the UQ analysis, including the estimation of local statistical moments and Sobol' indices, was performed. The obtained results indicate that both algorithms provide highly accurate approximations, even with a relatively small experimental design consisting of only 75 samples, when compared to the reference solution based on 1000 finite element model evaluations. Future work will focus on incorporating active learning strategy proposed by Novak et al. (2021) to further reduce the computational cost and on applying the entire methodology to complex three-dimensional case studies with very high-dimensional QoIs.

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